Linux Lab Report

Directories, Files, and Links Management

March 9, 2026

Part 1 – Exploring the File System

Commands Executed

Display the current directory:

```
mia@fedora:~$ pwd
/home/mia
```

List files and directories (ls):

```
mia@fedora:~$ ls
Desktop    Downloads  file2  myfile1  myfile3  myfile5  myfile7    Pictures  sourcefile  Videos
Documents  file1      Music  myfile2  myfile4  myfile6  myfiletxt  Public    Templates
mia@fedora:~$
```

List with details (ls -l):

```
mia@fedora:~$ ls -l
total 40
drwxr-xr-x. 1 mia mia 110 Jan 27 09:30 Desktop
drwxr-xr-x. 1 mia mia   0 Jan 12 11:08 Documents
drwxr-xr-x. 1 mia mia   0 Jan 12 17:28 Downloads
-rw-r--r--. 1 mia mia   0 Feb 10 08:58 file1
-rw-r--r--. 1 mia mia 151 Feb 16 08:57 file2
drwxr-xr-x. 1 mia mia   0 Jan 12 11:08 Music
-rw-r--r--. 1 mia mia 147 Feb 16 09:17 myfile1
-rw-r--r--. 1 mia mia  42 Feb 16 10:24 myfile2
-rw-r--r--. 1 mia mia   4 Feb 16 10:25 myfile3
-rw-r--r--. 1 mia mia  46 Feb 16 10:28 myfile4
-rw-r--r--. 1 mia mia 135 Feb 16 10:39 myfile5
-rw-r--r--. 1 mia mia   5 Feb 16 10:51 myfile6
-rw-r--r--. 1 mia mia   5 Feb 16 10:52 myfile7
-rw-r--r--. 1 mia mia 151 Mar  2 08:58 myfiletxt
drwxr-xr-x. 1 mia mia  56 Jan 13 09:46 Pictures
drwxr-xr-x. 1 mia mia   0 Jan 12 11:08 Public
-rw-r--r--. 1 mia mia 372 Mar  3 08:55 sourcefile
drwxr-xr-x. 1 mia mia   0 Jan 12 11:08 Templates
drwxr-xr-x. 1 mia mia  12 Jan 13 09:46 Videos
mia@fedora:~$
```

List all including hidden (ls -la):

```
mia@fedora:~$ ls -la
total 84
drwx------. 1 mia  mia   820 Mar  9 08:20 .
drwxr-xr-x. 1 root root    6 Jan 12 11:07 ..
-rw-------. 1 mia  mia   881 Feb 16 09:15 .bash_history
-rw-r--r--. 1 mia  mia    18 Jul 23  2025 .bash_logout
-rw-r--r--. 1 mia  mia   144 Jul 23  2025 .bash_profile
-rw-r--r--. 1 mia  mia   522 Jul 23  2025 .bashrc
drwx------. 1 mia  mia   470 Jan 13 09:08 .cache
drwxr-xr-x. 1 mia  mia   412 Jan 13 09:46 .config
drwxr-xr-x. 1 mia  mia   110 Jan 27 09:30 Desktop
drwxr-xr-x. 1 mia  mia     0 Jan 12 11:08 Documents
drwxr-xr-x. 1 mia  mia     0 Jan 12 17:28 Downloads
-rw-r--r--. 1 mia  mia     0 Feb 10 08:58 file1
-rw-r--r--. 1 mia  mia   151 Feb 16 08:57 file2
drwx------. 1 mia  mia    20 Jan 12 11:08 .local
drwxr-xr-x. 1 mia  mia    48 Jan 12 17:24 .mozilla
drwxr-xr-x. 1 mia  mia     0 Jan 12 11:08 Music
-rw-r--r--. 1 mia  mia   147 Feb 16 09:17 myfile1
-rw-r--r--. 1 mia  mia    42 Feb 16 10:24 myfile2
-rw-r--r--. 1 mia  mia     4 Feb 16 10:25 myfile3
-rw-r--r--. 1 mia  mia    46 Feb 16 10:28 myfile4
-rw-r--r--. 1 mia  mia   135 Feb 16 10:39 myfile5
-rw-r--r--. 1 mia  mia     5 Feb 16 10:51 myfile6
-rw-r--r--. 1 mia  mia     5 Feb 16 10:52 myfile7
-rw-r--r--. 1 mia  mia   151 Mar  2 08:58 myfiletxt
drwxr-xr-x. 1 mia  mia    56 Jan 13 09:46 Pictures
drwxr-xr-x. 1 mia  mia     0 Jan 12 11:08 Public
-rw-r--r--. 1 mia  mia   372 Mar  3 08:55 sourcefile
drwxr-xr-x. 1 mia  mia     0 Jan 12 11:08 Templates
-rw-r-----. 1 mia  mia     5 Mar  9 08:20 .vboxclient-clipboard-tty2-control.pid
-rw-r-----. 1 mia  mia     5 Mar  9 08:20 .vboxclient-clipboard-tty2-service.pid
-rw-r-----. 1 mia  mia     5 Mar  9 08:20 .vboxclient-draganddrop-tty2-control.pid
-rw-r-----. 1 mia  mia     5 Mar  9 08:20 .vboxclient-hostversion-tty2-control.pid
-rw-r-----. 1 mia  mia     5 Mar  9 08:20 .vboxclient-seamless-tty2-control.pid
drwxr-xr-x. 1 mia  mia    12 Jan 13 09:46 Videos
-rw-------. 1 mia  mia  4323 Feb 16 09:17 .viminfo
mia@fedora:~$
```

Questions

Q1.1 – What is the difference between ls and ls -l?

ls lists only the names of files and directories in the current folder.
ls -l (long format) shows detailed information for each entry, such as permissions, number of hard links, owner, group, file size in bytes, last modification date/time, and the name.

Q1.2 – What does ls -la display?

ls -la combines the long format (-l) with the show-all flag (-a).
It displays all files including hidden ones: those whose names start with a dot.
This reveals system and configuration files that ls and ls -l normally hide.

Part 2 – Directory Management

Commands Executed

Create and enter linux_lab directory:

```
mia@fedora:~$ mkdir linux_lab
mia@fedora:~$ cd linux_lab
mia@fedora:~/linux_lab$
```

Create sub-directories:

```
mia@fedora:~/linux_lab$ mkdir docs scripts backups
mia@fedora:~/linux_lab$
```

Create nested directories:

```
mia@fedora:~/linux_lab$ mkdir -p projects/project1/data
mia@fedora:~/linux_lab$
```

Display structure:

```
mia@fedora:~/linux_lab$ tree .
.
├── backups
├── docs
├── projects
│   └── project1
│       └── data
└── scripts

7 directories, 0 files
mia@fedora:~/linux_lab$
```

Remove scripts directory:

```
mia@fedora:~/linux_lab$ rmdir scripts
mia@fedora:~/linux_lab$
```

Remove projects directory (with contents):

```
mia@fedora:~/linux_lab$ rm -r projects
mia@fedora:~/linux_lab$
```

Final directory structure:

```
mia@fedora:~/linux_lab$ tree .
.
├── backups
└── docs

3 directories, 0 files
mia@fedora:~/linux_lab$
```

Questions

Q2.1 – What is the difference between rmdir and rm -r?

rmdir only removes empty directories. If the directory contains any files or sub directories it will refuse and print an error.
rm -r (recursive remove) deletes a directory and all its contents such as files, sub-directories, and everything nested inside, without warning.

Q2.2 – Why can rmdir fail?

rmdir fails when the target directory is not empty.
For example, if projects/ still contains project1/data/
rmdir projects will produce an error.
The solution is either to first remove all contents manually or to use rm -r instead.

Part 3 – Regular File Management

Commands Executed

Create two empty files:

```
mia@fedora:~/linux_lab$ touch file1.txt file2.txt
mia@fedora:~/linux_lab$
```

Write content to file1.txt:

```
mia@fedora:~/linux_lab$ echo "content file1" > file1.txt
mia@fedora:~/linux_lab$ cat file1.txt
content file1
mia@fedora:~/linux_lab$
```

Append content to file1.txt:

```
mia@fedora:~/linux_lab$ echo "appended content" >> file1.txt
mia@fedora:~/linux_lab$ cat file1.txt
content file1
appended content
mia@fedora:~/linux_lab$
```

Copy file1.txt to docs/:

```
mia@fedora:~/linux_lab$ cp file1.txt docs/
mia@fedora:~/linux_lab$
```

Rename file2.txt to notes.txt:

```
mia@fedora:~/linux_lab$ mv file2.txt notes.txt
mia@fedora:~/linux_lab$
```

Check metadata of file1.txt:

```
mia@fedora:~/linux_lab$ ls -li file1.txt
14995 -rw-r--r--. 1 mia mia 31 Mar  9 09:40 file1.txt
mia@fedora:~/linux_lab$
```

Part 4 – Hard Links

Commands Executed

Create a hard link:

```
mia@fedora:~/linux_lab$ ln file1.txt file1hlink.txt
mia@fedora:~/linux_lab$
```

Check inode numbers:

```
mia@fedora:~/linux_lab$ ls -li file1.txt file1hlink.txt
14995 -rw-r--r--. 2 mia mia 31 Mar  9 09:40 file1hlink.txt
14995 -rw-r--r--. 2 mia mia 31 Mar  9 09:40 file1.txt
mia@fedora:~/linux_lab$
```

Modify the link:

```
mia@fedora:~/linux_lab$ echo "modified via hlink" >> file1hlink.txt
mia@fedora:~/linux_lab$ cat file1.txt
content file1
appended content
modified via hlink
mia@fedora:~/linux_lab$ cat file1hlink.txt
content file1
appended content
modified via hlink
mia@fedora:~/linux_lab$
```

Delete the original file:

```
mia@fedora:~/linux_lab$ rm file1.txt
mia@fedora:~/linux_lab$ cat file1hlink.txt
content file1
appended content
modified via hlink
mia@fedora:~/linux_lab$ ls -li file1hlink.txt
14995 -rw-r--r--. 1 mia mia 50 Mar  9 09:49 file1hlink.txt
mia@fedora:~/linux_lab$
```

Questions

Q4.1 – What do you observe after modifying via the hard link?

Both file1.txt and file1hlink.txt show the new line when read with cat.
This confirms that a hard link is not a copy.
Both names point to the exact same inode 14995, and therefore the same data on disk.
Editing through either name changes the one shared data block.

Q4.2 – What do you observe after deleting the original file?

The content is still fully accessible through file1hlink.txt.
Deleting file1.txt only removed that directory entry (name), not the data.
The inode's link count dropped from 2 to 1, but the data remains intact because at least one hard link still references it.
The file is only truly deleted when its link count reaches 0.

Part 5 – Symbolic Links

Commands Executed

Create a symbolic link to “hardlink_file1.txt”:

```
mia@fedora:~/linux_lab$ ln -s file1hlink.txt file1sym.txt
mia@fedora:~/linux_lab$
```

Display link metadata:

```
mia@fedora:~/linux_lab$ ls -li file1sym.txt
14998 lrwxrwxrwx. 1 mia mia 14 Mar  9 09:53 file1sym.txt -> file1hlink.txt
mia@fedora:~/linux_lab$
```

Delete the target file:

```
mia@fedora:~/linux_lab$ rm file1hlink.txt
mia@fedora:~/linux_lab$ ls -li file1sym.txt
14998 lrwxrwxrwx. 1 mia mia 14 Mar  9 09:53 file1sym.txt -> file1hlink.txt
mia@fedora:~/linux_lab$ cat file1sym.txt
cat: file1sym.txt: No such file or directory
mia@fedora:~/linux_lab$
```

Questions

Q5.1 – What do you observe after deleting the target?

The symbolic link still exists as a file, but it becomes broken link.
Unlike a hard link, a symbolic link stores only the path to the target, it has no direct connection to the inode.
Once the target is deleted, the path it stores leads nowhere, so any attempt to read the symbolic link fails with "No such file or directory".

Final Challenge

Commands Executed

Create linux_lab_final/ directory: Create reports/archive directories:

```
mia@fedora:~/linux_lab$ pwd
/home/mia/linux_lab
mia@fedora:~/linux_lab$ cd
mia@fedora:~$ pwd
/home/mia
mia@fedora:~$ mkdir linux_lab_final
mia@fedora:~$ cd linux_lab_final
mia@fedora:~/linux_lab_final$ pwd
/home/mia/linux_lab_final
mia@fedora:~/linux_lab_final$
```

Create reports/archive directories:

```
mia@fedora:~/linux_lab_final$ mkdir reports archive
mia@fedora:~/linux_lab_final$ tree .
.
├── archive
└── reports

3 directories, 0 files
mia@fedora:~/linux_lab_final$
```

Create the report.txt file:

```
mia@fedora:~/linux_lab_final$ echo "final report" > reports/report.txt
mia@fedora:~/linux_lab_final$ tree .
.
├── archive
└── reports
    └── report.txt

3 directories, 1 file
mia@fedora:~/linux_lab_final$
```

Create hard link:

```
mia@fedora:~/linux_lab_final$ ln reports/report.txt reporthlink.txt
mia@fedora:~/linux_lab_final$ ls -li reports/report.txt reporthlink.txt
15016 -rw-r--r--. 2 mia mia 13 Mar  9 10:18 reporthlink.txt
15016 -rw-r--r--. 2 mia mia 13 Mar  9 10:18 reports/report.txt
mia@fedora:~/linux_lab_final$
```

Create symbolic link:

```
mia@fedora:~/linux_lab_final$ ln -s reports/report.txt reportsym.txt
mia@fedora:~/linux_lab_final$ ls -li reports/report.txt reportsym.txt
15016 -rw-r--r--. 2 mia mia 13 Mar  9 10:18 reports/report.txt
15021 lrwxrwxrwx. 1 mia mia 18 Mar  9 10:27 reportsym.txt -> reports/report.txt
mia@fedora:~/linux_lab_final$
```

Verify directory structure:

```
mia@fedora:~/linux_lab_final$ tree .
.
├── archive
├── reporthlink.txt
├── reports
│   └── report.txt
└── reportsym.txt -> reports/report.txt

3 directories, 3 files
mia@fedora:~/linux_lab_final$
```

Verify inode numbers (ls -li):

```
mia@fedora:~/linux_lab_final$ ls -li
total 8
15015 drwxr-xr-x. 1 mia mia  0 Mar  9 10:13 archive
15016 -rw-r--r--. 2 mia mia 13 Mar  9 10:18 reporthlink.txt
15014 drwxr-xr-x. 1 mia mia 20 Mar  9 10:18 reports
15021 lrwxrwxrwx. 1 mia mia 18 Mar  9 10:27 reportsym.txt -> reports/report.txt
mia@fedora:~/linux_lab_final$ ls -li reports/
total 4
15016 -rw-r--r--. 2 mia mia 13 Mar  9 10:18 report.txt
mia@fedora:~/linux_lab_final$
```

Analysis:

report.txt (inode 15016) and reporthlink.txt (inode 15016) share the same inode number, confirming they are two directory entries pointing to the same data block.
The link count is 2. reportsym.txt (inode 15021) has a different inode, it is an independent file that stores only the path "reports/report.txt", not a direct reference to the data.